deploy it on a production blockchain, because when the smart contract is deployed, it is immutable.

Your goal in the design process is to define the contents of a smart contract; specifically, its

- Data
- Functions that operate on the data
- Rules for operations

Design principle 2 initiates the design process by defining the users of the system that will be served by the application.

> **DESIGN PRINCIPLE 2** Define the users of and use cases for the system. Users are entities that generate the actions and the input and receive the output from the system you'll be designing.

2.2.1 A use case diagram for the counter

Let's apply these principles and begin the design process for the counter problem with the standard Unified Modeling Language (UML; https://www.uml.org/ index.htm) tools for design representation: the use case diagram and the class diagram. The UML design representation may be familiar to many advanced developers, but a beginner developer may not be familiar with this. You can refer to the UML description in appendix A to create these diagrams.

The UML use case diagram helps you think through the problem and decide how the smart contract--and, more specifically, its functions--will be used. Figure 2.3 shows just